Chapter 9

To the Information Storehouse

You might see various kinds of information about the match, the players, and the venue being displayed on television during the live telecast of sports events. Have you ever wondered how such interesting and contextual information becomes instantly available during the game?

All this information related to the matches is made available to us with the help of a pre-collected and stored database system.

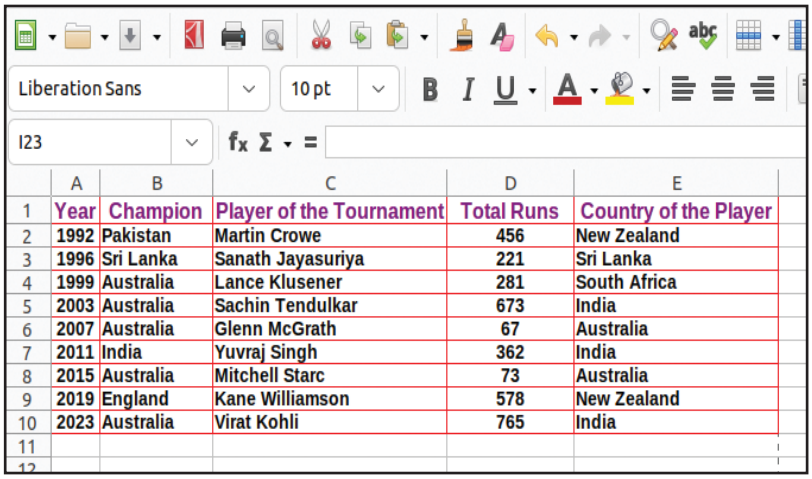
Knowingly or unknowingly, database has a major influence on our lives. A few examples are given below. Complete the ones that are incomplete:.

- Election results and related details displayed on television.
- Exam results and their analysis made available on websites.
- .....
- .....
- .....

You have learned in the previous class that spreadsheet software can be used to store and analyze information in tables.

Let's Find the Information in the List

Look at Figure 9.1. The table has been prepared in LibreOffice Calc with some information related to the Cricket World Cup.



Year	Champion	Player of the Tournament	Total Runs	Country of the Player
1992	Pakistan	Martin Crowe	456	New Zealand
1996	Sri Lanka	Sanath Jayasuriya	221	Sri Lanka
1999	Australia	Lance Klusener	281	South Africa
2003	Australia	Sachin Tendulkar	673	India
2007	Australia	Glenn McGrath	67	Australia
2011	India	Yuvraj Singh	362	India
2015	Australia	Mitchell Starc	73	Australia
2019	England	Kane Williamson	578	New Zealand
2023	Australia	Virat Kohli	765	India

Fig. 9.1 Table Prepared in LibreOffice Calc

Can you find the answers to the following questions from this table?

- Which country won the Cricket World Cup in 2003?
  - ➢ The best player in the tournament
  - ➢ His country
- How many times has the player from the country that won the Cricket World Cup been the best player in that tournament?

All this information can be made available very quickly on examining the data in the table given above. Here, we need to examine only a few data. But if we examine the data of all the cricket matches played since the first official cricket test between England and Australia in 1877, is it easy to examine and find the information of thousands of matches and as many players from one table?

Try to find out the answer of the following question too from the figure 9.1

- Which New Zealanders have been the tournament's best players?

Martin Crowe, Kane Williamson etc., right?

Information from Different Lists

Look at the information in the spreadsheet mentioned above, recorded in another form as tables.

Country Table

Country_ID	Country_Name
C100	Pakistan
C101	Sri Lanka
C102	India
C103	Australia
C104	New Zealand
C105	South Africa
C106	England

Table 9.1 Details of the country

Player Table

Player_ID	Player_Name	Total_Runs	Country_ID
P5001	Sanath Jayasurya	221	C101
P5002	Sachin Tendulkar	673	C102
P5003	Lance Clusener	281	C105
P5004	Martin Crowe	456	C104
P5005	Glenn McGrath	67	C103
P5006	Yuvaraj Singh	362	C102
P5007	Mitchel Starc	73	C103
P5008	Kane Williamson	578	C104
P5009	Virat Kohli	765	C102

Table 9.1 Details of the players

Matches Table

Year	Winner	Best Player
1992	C100	P5004
1996	C101	P5001
1999	C103	P5003
2003	C103	P5002
2007	C103	P5005
2011	C102	P5006
2015	C103	P5007
2019	C106	P5008
2023	C103	P5009

Table 9.1 Details of the Matches

Have you examined all the tables?

Are the answers to the questions we discussed earlier available from these tables?

For that, we will have to relate the three tables and examine them, right?

From these new tables, is it possible to find out from which country is the best player of 2011 World Cup?

How could you arrive at this answer?

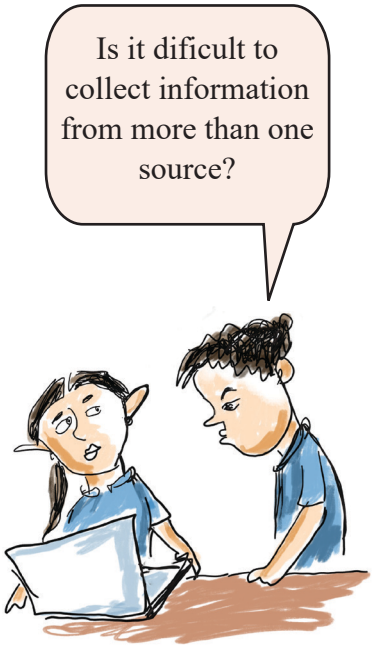
- In the new tables, a special indicator (ID) has been given to the country, players, winner, and best player. Based on this,
  - We can find the ID of the best player in the 2011 World Cup from the Matches Table. This ID is P5006.
  - We can understand from the Player Table that the ID P5006 belongs to the player Yuvraj Singh.
  - We can find out from the Player Table itself that Yuvraj Singh's Country_ID is C102.
  - It is given in the Country Table that C102 is the Country_ID of India.

That is, the best player in the 2011 World Cup is an Indian.

Thus, all the information in the Calc sheet shown at the beginning of the lesson can be found by relating these three tables.

Is there any advantage in organizing the information in different tables like this?

- The same information is not recorded repeatedly in these tables.
- All the information related to a subject is put in one table here. You can understand their contents from the names of these tables. For example, Country, Player, etc.



To the Database Management System

Finding information from multiple tables by relating them is not as easy as finding information from one table.

But what if we had computer software that could help us find and provide the information we need from such interconnected

tables? Such software is called Database Management Systems (DBMS).

DBMSs not only provide us with the recorded information in the way we need, but also help us store it, make changes to it, and eliminate unnecessary ones. It is with the help of such software that we can display and analyze large amounts of collected information very quickly today.

DBMS Software

The information stored in DBMSs is presented to the user in an easy and attractive manner with the help of techniques like html,, Java, Python, etc. Important DBMS softwares are Oracle, MySQL, SQL Server, and sqlite.

DATABASE

We have examined the information from the tables related to the Cricket World Cup here. A database is a collection of tables in which such interrelated information are recorded in a systematic manner.

We can prepare a database by including information in this table in a DBMS software.

Suppose that we used a database named cricket_worldcup and its tables Country and Player to find the answers to the questions. We shall now prepare a database of this type and its tables.

Fig. 9.2 Tables of the Cricket_worldcup Database

See the tables 9.1 to 9.3. There are many rows and columns in each one of these tables. What are the columns in the table Player ?

- Player_ID
- Player_Name
- Total_Runs
- Country_ID

These are called Fields of the table in relation to Databases (Fig 9.3)

Player Table

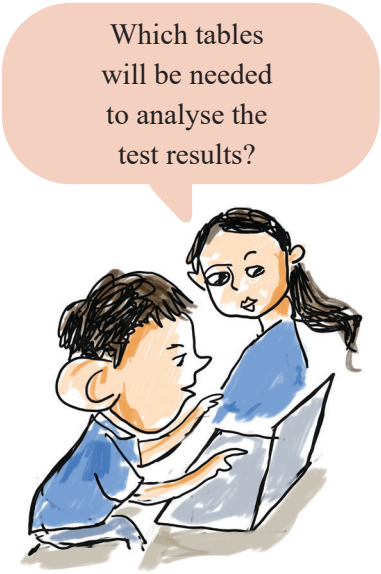
Fields

Player_ID	Player_Name	Total_Runs	Country_ID
P5001	Sanath Jayasurya	221	C101
P5002	Sachin Tendulkar	673	C102
P5003	Lance Clusener	281	C105
P5004	Martin Crowe	456	C104
P5005	Glenn McGrath	67	C103

Fig. 9.3 Fields of the Table

Similarly, note down the fields of the other tables (9.1, 9.3)

- Country : .....
- Matches : .....



The information collected in each field of the table is different. The names of the players contain letters whereas the field of Total_Runs will be numbers, right? That is, the information collected in each field will be composed of different Datatypes.

Observe the table given below.

Sl. No.	Fields	Datatype
1	ID of the country	Text
2	Year of game	Number
3	Date of game	Date
4	Picture of the player	Image

Table 9.4 Some Datatypes

You have seen some common datatypes. Apart from these, Audio and Video are also datatypes. But they are known by different names in each DBMS.

Details about the player is given in rows. These rows in the table are referred to as Records (Figure 9.4). All the information collected about a player can be found in a record in the Player table.

Records

Player Table

Player_ID	Player_Name	Total_Runs	Country_ID
P5001	Sanath Jayasurya	221	C101
P5002	Sachin Tendulkar	673	C102
P5003	Lance Clusener	281	C105
P5004	Martin Crowe	456	C104
P5005	Glenn McGrath	67	C103

Fig. 9.4 Records in the Table

In the Case of the Same Data

Look at Figure 9.3. Why is the field Player_ID included in this table?

Suppose there are more than one children with the same name in your class. Do you know what is used to identify such children with same name in the records related to the class?

- Their Roll No will be different, right?

Similarly, accurate data can be obtained only if we are able to distinguish each record in a table from the other records.

Thus, the fields added to identify each record in each table from others are called Primary key (Figure 9.5).

Primary Key

Player Table

Player_ID	Player_Name	Total_Runs	Country_ID
P5001	Sanath Jayasurya	221	C101
P5002	Sachin Tendulkar	673	C102
P5003	Lance Clusener	281	C105
P5004	Martin Crowe	456	C104
P5005	Glenn McGrath	67	C103

Fig. 9.5 Primary Key in the Table

Some other such cases where one record is added to distinguish one from another while storing information are given below. Try to add similar ones.

- Admission Number is given to students.
- Book code is given to books in the school library.
- School Code is given to all schools.
- .....
- .....
- .....

Let's Create a Database for the Cricket World Cup

We have already learned some things about database management systems. Let's see how to create a database containing information related to Cricket_worldcup matches using these. We shall do this using SQLite, a DBMS that belongs to the free software category.

Open the software DB Browser for SQLite on your computer and create a database with Cricket_worldcup as file name and save it in your folder.

To Prepare Database Using Sqlite

- Open DB Browser for SQLite on your computer.
- Click on New Database to create a new database (fig 9.6)
- Give the name of the Database (Cricket_worldcup) against Name in the window that opens.
- Then, select the folder to which you want to save and click Save.

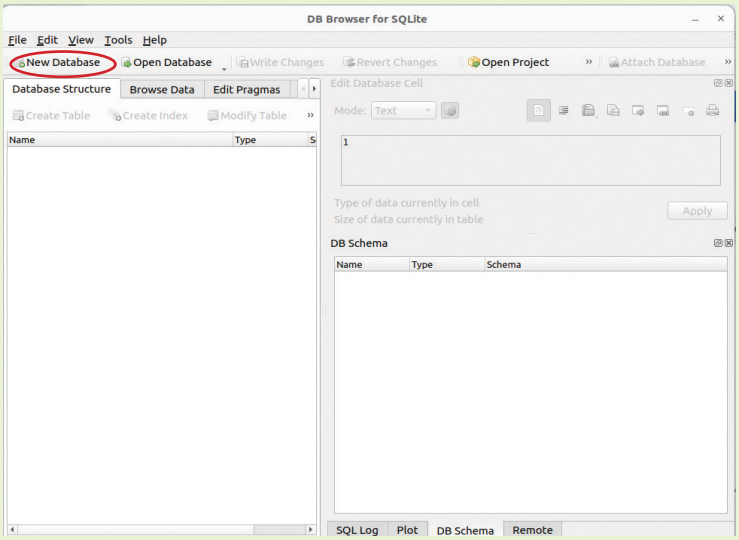


Fig. 9.6 DB Browser for SQLite Window

Thus, the window for creating table will be displayed. le creation window. We have to create our tables here.

Shall we prepare the table Country first?

To Prepare Table in DB Browser for SQLite

- Give the name of the table (Country) against Table in the Edit table definition window.
- Then, click on the Add button below.
- Edit the name of the field as Field1 against Name.
- Add other fields in the same way.

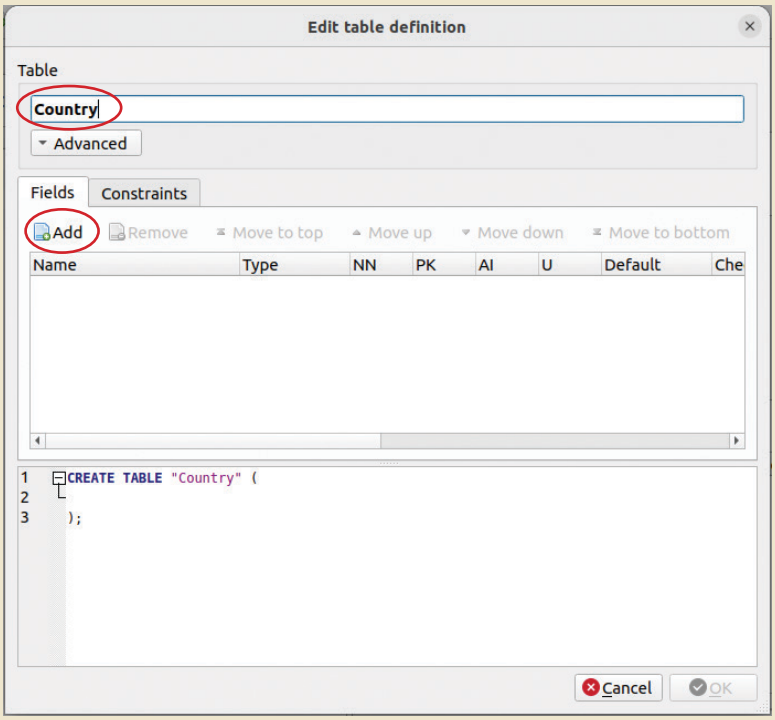


Fig. 9.7 DB Browser for SQLite-Edit Table Definition Window

Sqlite

Sqlite is a DBMS that is completely a free software. This system can be used in desktop software, mobile apps, and web applications. In Sqlite, the database is stored as a single file on disk. Therefore, the database can be easily managed and configured.

Sqlite was developed by D.Richard Hipp, a software engineer.

In the table Country, we need to add two fields - Country_ID and Country_Name. Once Country_ID is added, we can select its datatype.

Country_ID consists of letters and numbers, such as C100. So, guess what will be its datatype.

In our record, Country_ID is the field with a unique value used to identify it. Therefore, we can set Country_ID as the Primary Key.

To Set Primary Key in DB Browser for SQLite

- Click on the field which has to be set as the Primary Key
- Then, enable tick mark in PK.

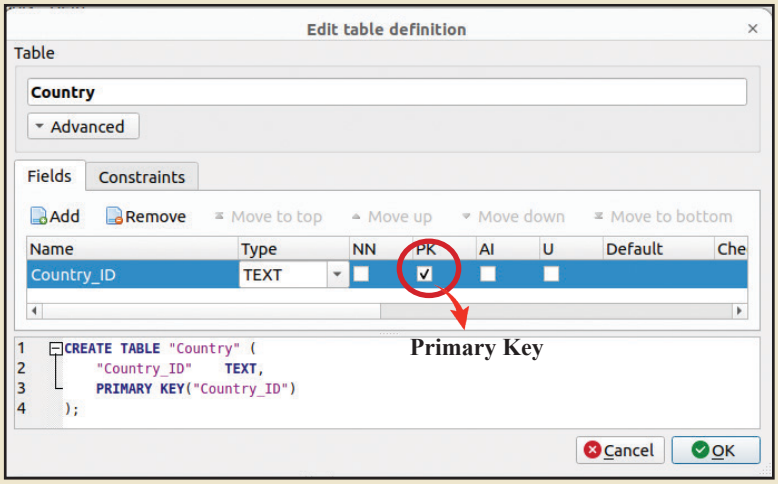


Fig. 9.8 How to set Primary Key in DB Browser for Sqite

- Click OK after including necessary fields.

Now, include the field Country_Name in the same way (Fig 9.9).

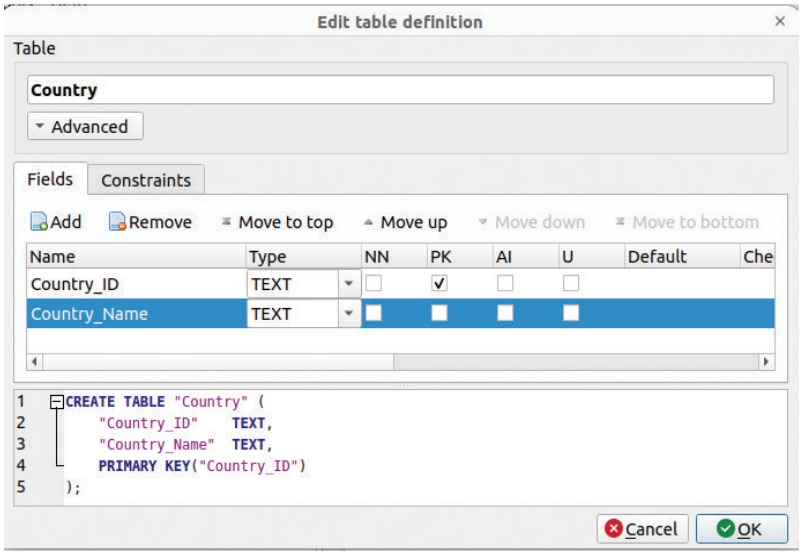


Fig. 9.9 Fields Included in Table

With this, you can see the table named Country under the Tables section of the main window (Figure 9.10).

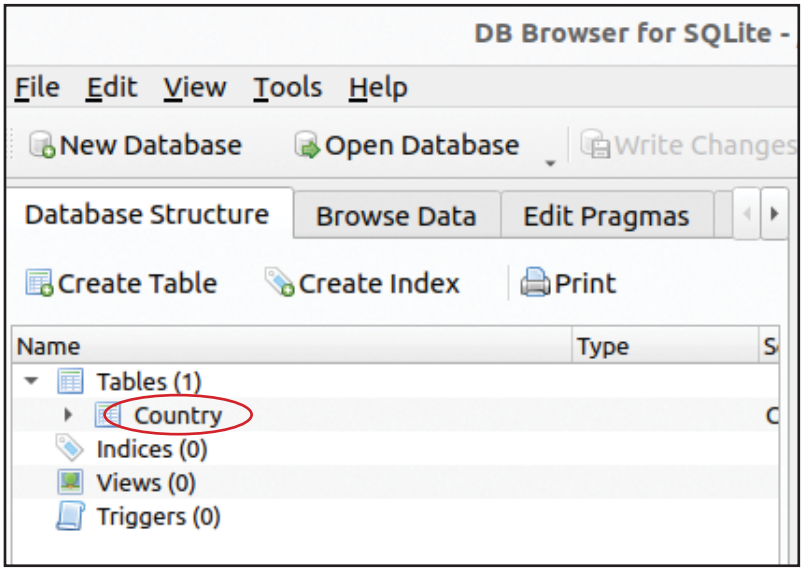


Fig. 9.10 DB Browser for SQLite – Database Structure window

Let's Add Information into Database

We have already created the required tables in the database related to the Cricket World Cup. Now we need to add information to these tables.

For this, Insert new record to current table option is available in the DB Browser for SQLite window.

To Add Records in DB Browser for SQLite

- Select the tab Browse Data in window.
- Select the table to which records are to be included. (Here we have created only Table, Country and so it will be selected)
- Then, click on the button Insert new record to current table.
- Now, add the records one by one by typing.

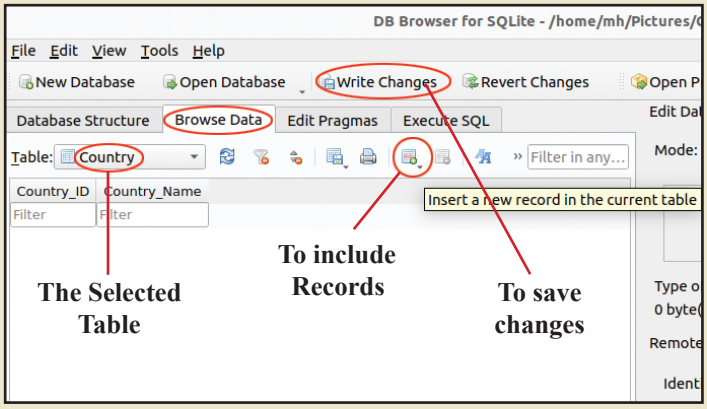


Fig 9.11 DB Browser for SQLite – Method to Add Records

In the table, Country that you have created, add the details from Table 9.1, one by one using the Insert new record to current table feature (Figure 9.12).

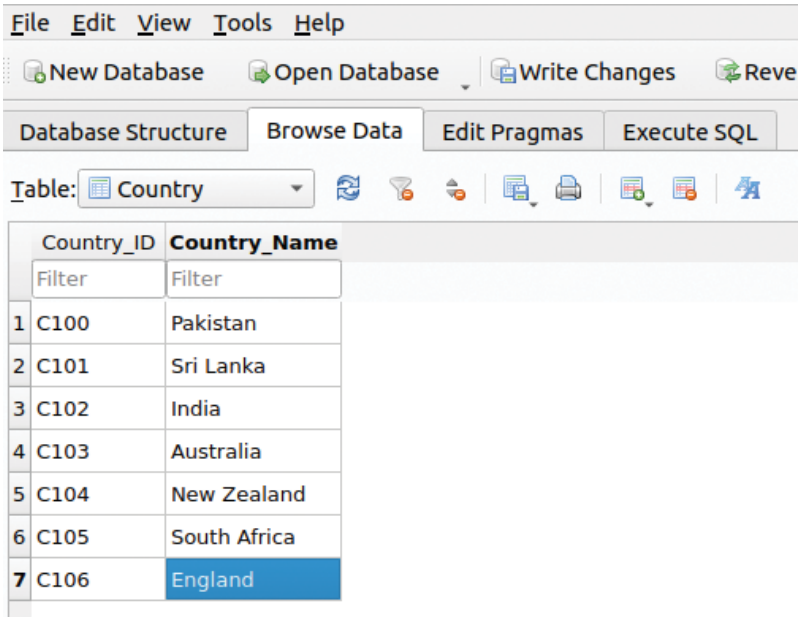


Fig. 9.12 Details Included in the Table Country

Similarly, prepare the tables – Player, Matches and include the details given in table 9.2 and 9.3 accordingly.

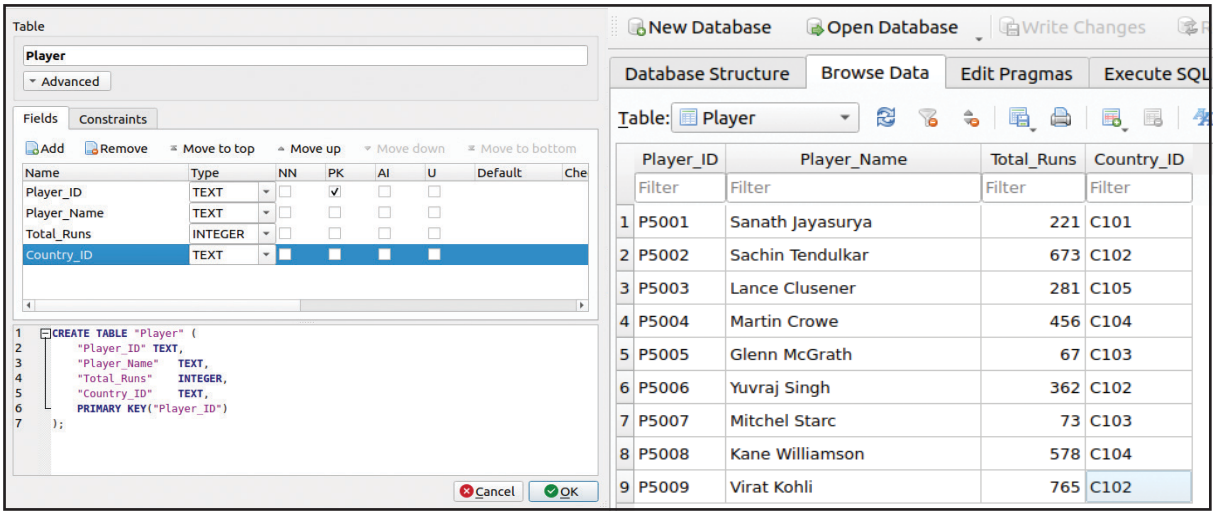


Fig. 9.13 Records Included in the Table Player

Discuss why INTEGER datatype has been given to the Total_Runs field of the table Player.

Let's Retrieve Information from the Database

We have created the tables required for the Cricket_Worldcup database and added various records to them. To view all the

records added to these tables, just click the Browse Data tab and select the table.

But how do we find only the information we need from this?

For example, suppose we want to find the number of runs Sachin Tendulkar has scored from this database.

There are only very few information about the players in Player table of the database we have created. So, we can open the table through the Browse Data tab and quickly find it. But if there are thousands of players in this table, is this task easy?

The way to do this is by using SQL Queries.

Notice the equivalent SQL queries are displayed when the table is created in the DB Browser (Figure 9.13).

So, to find the number of runs scored by Sachin Tendulkar from the Player table, simply enter the following query in the Execute SQL tab and run it (Figure 9.14).

'select Total_Runs from Player where Player_Name='Sachin Tendulkar'

Structured Query Language (SQL)

DBMSs communicate using a special computer language called Structured Query Language (SQL). It is a language that can be used very easily, like the English language. Not only can you select information from tables, but you can also create and modify tables, insert information into tables, edit and delete information.

In some DBMSs, these operations can only be performed using SQL.

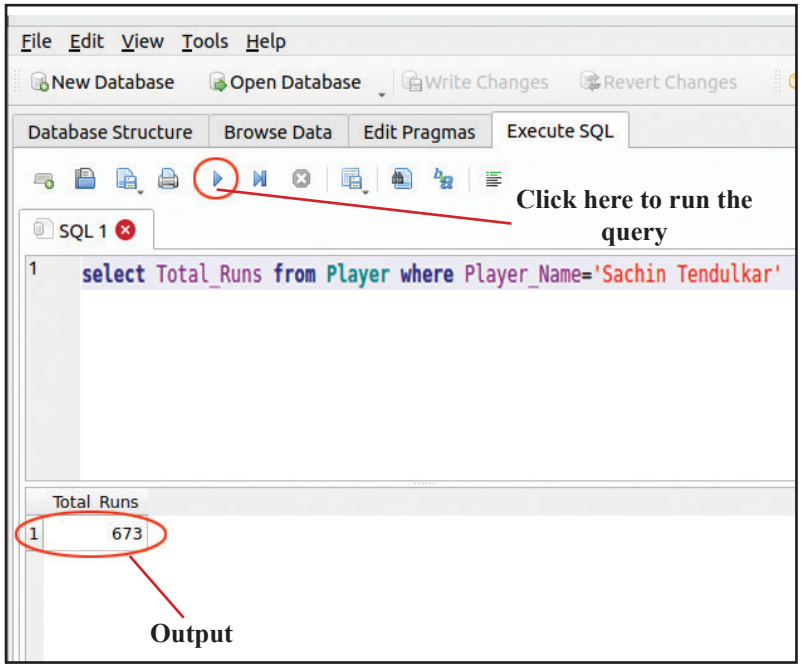


Fig. 9.14 Execute SQL Window

Now, write and run an SQL query to display the name of the player who scored 765 runs in the Player table.

- *select Player_Name from Player where Total_Runs=765*

More SQL Queries

Some SQL statements to retrieve information from the Player table are given below. Run them and complete the table.

	SQL Statement	Use
1	select Player_Name from Player where Total_Runs>300	to get the names of all the players who have scored more than 300 runs
2	select Player_Name from Player where Total_Runs between 200 and 400	
3	select Player_ID, Player_Name from Player where Country_ID='C103'	
4	select * from Player	to get all the details from the Player table
5	select Player_ID, Player_Name from Player where Country_ID in ('C103', 'C105')	
6		To get the names of the players who have scored less than 400 runs.

Table 9.5 Some SQL Statements and their Uses

Retrieving Data from Multiple Tables

We have now seen a few SQL queries that retrieve data from the Player table.

Now consider the query given below.

- Which country is the player 'Sanath Jayasuriya' from?

The Player table does not provide the country of each player. Instead, Country_ID is provided. To find the country with the given ID, we will have to examine multiple tables (including the Country_ID field in the Country table).

That is, here the field Country_ID in the Player table is used to find the relation with the Country table. Such fields are known as Foreign keys (Figure 9.15).

Player Table

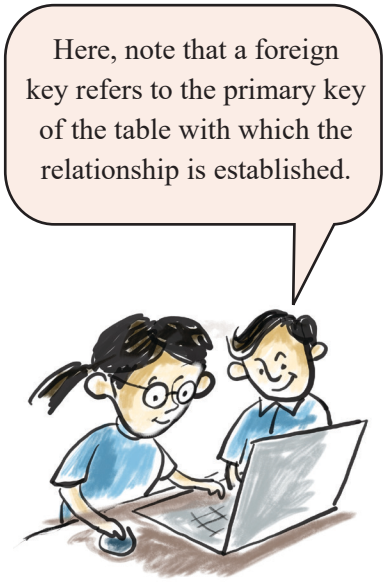
Player_ID	Player_Name	Total_Runs	Country_ID
P5001	Sanath Jayasurya	221	C101
P5002	Sachin Tendulkar	673	C102
P5003	Lance Clusener	281	C105
P5004	Martin Crowe	456	C104
P5005	Glenn McGrath	67	C103

Country Table

Country_ID	Country_Name
C100	Pakistan
C101	Sri Lanka
C102	India
C103	Australia
C104	New Zealand
C105	South Africa
C106	England

Foreign key

Fig. 9.15 When more than one Tables are Examined for Information



If so, what will be the SQL query to find out the nationality of 'Sanath Jayasurya' ?

We have discussed that this information will not be available from a single table. So,

- We get the Country_ID of Sanath Jayasurya as C101 if we give 0*Select Country_ID from Player where Player_Name='Sanath Jayasurya' as query.*
- We can know that the country with C101 as ID is Sri Lanka from the Country table. But if there are hundreds of countries in this table, this is not easy.

If so, what will be the query to find out the name of the country with C101 as ID from the Country table?

- *select Country_Name from Country where Country_ID='C101'*

Now, combine this statement with the former one -

- *select Country_Name from Country where Country_ID=(Select Country_ID from Player where Player_Name='Sanath Jayasurya')*

In the same way, try writing and running an SQL statement to find out which country the player Lance Clusener belongs to.

Let's Asses

- Which one is used to distinguish the records of a table from the others?

  a) primary key b) foreign key c) fields d) private key

- SQL means .....?

  a) Systematic Query Language b) Structured Query Language
  c) Structured Questions Language d) Sequenced Query Language

Extended Activities

1. We have understood the need to control internet usage and screen time. Below is a table containing the data collected from a survey conducted among students regarding screen time.

   Create a database called Screen_Usage and prepare the tables as shown below.

App_ID	App_Name	App_Category	Usage_Duration (Minutes)
100	Samagra AI	Education	60
101	Facebook	Social Media	32
102	Youtube	Social Media	15
103	E-cube English	Education	25
104	G-Combris	Education	20
105	Whatsapp	Messaging	35
106	G-pay	Banking	10
107	Maps	Navigation	5

Now, try writing and executing the following SQL statements from the Apps table.

- Which all Educational Apps are made available?
- Which all Apps are being used more than 30 minutes in a day?
- To which category does the SAMAGRA AI App belong?